

Artery Geometry Extraction: Hybrid Approach

Phase 1: Pre-processing & RCA/LCA Separation

1. **Load the unified mask:** Read the `.nrrd` file and extract the raw 3D NumPy array.
2. **Connected Components:** Use `scipy.ndimage.label` to identify the distinct, unconnected objects in the 3D grid. The two largest objects will be the RCA and LCA.
3. **Center of Mass Sorting:** Implement Maren's logic to calculate the spatial center of each object. The one furthest to the right (lower X coordinate in standard medical spacing) is the RCA; the one furthest to the left is the LCA.
4. **Split the Data:** Create two separate binary arrays (one for RCA, one for LCA).

Phase 2: The Loop (Process RCA, then process LCA)

From here, we put the following steps inside a loop so the exact same logic runs for both the RCA and the LCA individually.

Step 2A: Mesh Generation & Smoothing

- Convert the separated numpy array back into a VMTK `vtkImageData` object.
- Run `vtkMarchingCubes` to create the surface.
- Run `vtkSurfaceSmoothing` (with our optimized **20 iterations**) to get the continuous mathematical mesh.

Step 2B: Automated Seed Point Extraction (Maren's "Scout")

- Run `skimage.morphology.skeletonize_3d` on the separated binary array.
- Scan the resulting skeleton for **endpoints** (voxels with exactly 1 neighbor).
- Compute the Distance Transform of the binary mask to find the thickness at each endpoint.
- Assign the endpoint with the largest thickness as the **Source** (Ostium). Assign all other endpoints as **Targets** (Distal branches).
- Convert these voxel indices (`z, y, x`) into physical coordinates (`X, Y, Z`) in millimeters using the Origin and Spacing metadata.
- Snap these coordinates to the nearest vertex on the smoothed mesh.

Step 2C: VMTK Mathematical Centerline Extraction

- Feed the smoothed mesh, the **Source** coordinate, and the **Target** coordinates into `vtkscripts.vtkCenterlines`.
- Execute the algorithm to trace the mathematically optimal geometric centerlines.

Phase 3: Data Extraction & DataFrame Construction

1. Extract the raw arrays from the VMTK centerline output. Specifically, pull the **Points** array (`Px, Py, Pz`) and the **MaximumInscribedSphereRadius** array.
2. Create a temporary Pandas DataFrame for the current artery containing:
 - **Sample_ID:** The identifier you passed at the start (e.g., "Normal_1").
 - **Artery_Type:** The current loop variable ("RCA" or "LCA").
 - **Px, Py, Pz:** The physical coordinates of the centerline.
 - **Radius:** The inscribed radius at that exact point.

3. Append this temporary DataFrame to a "Master" DataFrame.

Why does this architecture work?

By combining Maren's separation and skeletonization logic with VMTK's exact geometry extraction, the entire pipeline becomes 100% automated. You could point this script at a folder of 100 ASOCA patients, hit "Run", go grab a coffee, and come back to a massive, perfectly formatted CSV file ready for stenosis quantification.